CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 4: **Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $x + \frac{1}{x} = 4$, then $x^3 + \frac{1}{x^3} =$ 1

(a) 51 **(b)** 52 **(c)** 54 **(d)** 53

2. The value of 99^3 is: 1

(a) 970298 **(b)** 970300 **(c)** 970299 **(d)** 970301

3. If $x + y = 9$ and $x^2 + y^2 = 53$, then $xy =$ 1

(a) 15 **(b)** 14 **(c)** 16 **(d)** 13

4. If $x - y = 5$ and $x^2 + y^2 = 73$, then $xy =$ 1

(a) 26 **(b)** 24 **(c)** 23 **(d)** 25

5. If $x^2 + \frac{1}{x^2} = 23$, then $\left(x + \frac{1}{x}\right)^2 =$ 1

(a) 24 **(b)** 25 **(c)** 26 **(d)** 27

6. $27x^3 - 8$ factorises as: 1

(a) $(3x - 2)(9x^2 + 6x + 4)$ **(b)** $(3x - 2)(9x^2 - 6x + 4)$ **(c)** $(3x - 2)^3$ **(d)** $(3x + 2)(9x^2 - 6x + 4)$

7. If $x + y = 6$ and $xy = 8$, then $x^3 + y^3 =$ 1

(a) 74 **(b)** 72 **(c)** 71 **(d)** 73

8. If $x - y = 1$ and $xy = 12$, then $x^3 - y^3 =$ 1

(a) 38 **(b)** 36 **(c)** 37 **(d)** 39

9. **Assertion (A):** $(x - y)^2 = x^2 - 2xy + y^2$. 1
Reason (R): $(x - y)^2 = x^2 - y^2$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** $(a+b)^3 = a^3 + b^3$ for all real numbers a and b. 1

Reason (R): $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the value of 97^3 using an identity. 2

12. Factorise $27x^3 + 8$. 2

13. If $x + y = 6$ and $xy = 5$, find $x^3 + y^3$. 2

14. Find the value of 105^3 using an identity. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If $x + \frac{1}{x} = 9$, find $x^3 + \frac{1}{x^3}$. 3

16. Factorise $x^2 - 8x + 15$ and use it to find the value when $x = 9$. 3

17. Factorise $x^3 - 216$. 3

OR

If $x + \frac{1}{x} = 10$, find $x^3 + \frac{1}{x^3}$. 3

18. Factorise $x^2 - 11x + 18$ and use it to find the value when $x = 12$. 3

OR

Factorise $125x^3 - 8$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A square lawn of side x metres is surrounded by a path 2 m wide on all four sides, so that the outer boundary is also a square. 5

(i) Write the side of the outer square.

(ii) Expand the area of the outer square.

(iii) Find the area of the path in terms of x.

(iv) If $x = 12$, find the area of the path and the cost of paving it at Rs 50 per m^2.

20. If $x - y = 4$ and $xy = 45$, find $x^2 + y^2$, $x^3 - y^3$ and $(x+y)^2$. 5

OR

If $x + y = 13$ and $x^2 + y^2 = 89$, find xy, $(x-y)^2$ and $x^2 - y^2$ given that $x > y$. 5

END OF QUESTION PAPER — **Total: 40 Marks**